

Hints & Solutions

#MathBoleTohMathonGo

Q1 (2) - Single Correct

$$\text{Given, } \log_x y = -\frac{1}{4} \Rightarrow y = x^{-1/4} \Rightarrow x^{1/4} y = 1$$

$$\Rightarrow xy^4 = 1 \quad \dots(i)$$

$$\text{and } E = \log_x (xy^5) - \left[ 2 \log_y x - \frac{1}{2} \log_y y \right]$$

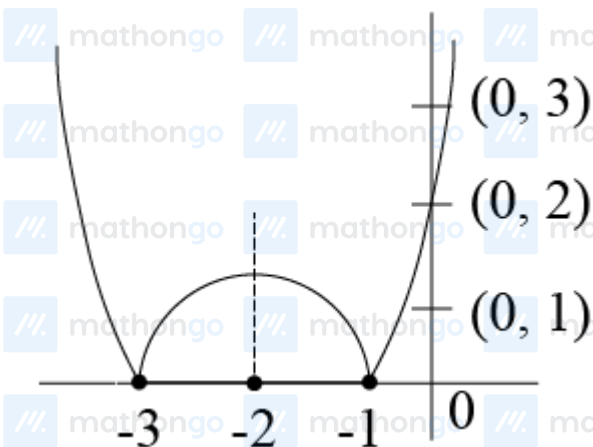
$$= \log_x (xy^4 \cdot y) - 2 \log_y x + \frac{1}{2}$$

$$= \log_x (xy^4) + \log_x y - 2 \log_y x + \frac{1}{2}$$

$$= 0 + \left(-\frac{1}{4}\right) - 2(-4) + \frac{1}{2} = -\frac{1}{4} + 8 + \frac{1}{2} = 8 + \frac{1}{4} = \frac{33}{4}$$

Q2 (4) - Single Correct

Let  $f(x) = |(x+1)(x+3)|$  shown as below



For 3 solutions,  $y = mx - 2m$  must be a tangent when  $x \in (-3, -1) \therefore y = -(x^2 + 4x + 3)$  has tangent

$$y = mx - 2m, \text{ when } x \in (-3, -1)$$

$$\Rightarrow x^2 + 4x + 3 + mx - 2m = 0$$

$$\Rightarrow x^2 + x(4+m) + (3-2m) = 0$$

has equal roots.

$$\therefore D = 0$$

$$\Rightarrow (4+m)^2 - 4(3-2m) = 0$$

$$\Rightarrow 16 + 8m + m^2 - 12 + 8m = 0$$

$$\Rightarrow m^2 + 16m + 4 = 0$$

$$\Rightarrow m = -8 \pm \sqrt{60}$$

Q3 (1) - Single Correct

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$$\text{Let } x^2 - 5x + 4 = a \quad \dots(i)$$

$$\therefore a(y^2 + y + 1) < 2y$$

$$\Rightarrow ay^2 + y(a - 2) + a < 0 \text{ for all } y \in R$$

$$\therefore a < 0 \text{ and } D < 0$$

$$\Rightarrow a < 0 \text{ and } (a - 2)^2 - 4a^2 < 0$$

$$\Rightarrow 3a^2 + 4a - 4 > 0$$

$$\therefore a < 0 \text{ and } a < -2$$

$$\Rightarrow a < -2 \quad \dots(ii)$$

$$\text{From Eqs. (i) and (ii), } x^2 - 5x + 4 < -2$$

$$\Rightarrow x^2 - 5x + 6 < 0 \Rightarrow x \in (2, 3)$$

**Q4 (1) - Single Correct**

$$x^4 + 3 \cos(ax^2 + bx + c) = 2(x^2 - 2)$$

$$\Rightarrow (x^2 - 1)^2 = -3 \{1 + \cos(ax^2 + bx + c)\}$$

Solution exists when LHS = RHS = 0

$$\Rightarrow x = \pm 1 \text{ and } \cos(ax^2 + bx + c) = -1$$

When  $x = 1$ , then

$$\cos(a + b + c) = -1$$

When  $x = -1$  then

$$\cos(a - b + c) = -1$$

As  $(a, b, c) \in (2, 5)$

$$\Rightarrow a + b + c \in (6, 15)$$

and  $a - b + c \in (-1, 8)$

$$\therefore a + b + c = 3\pi \text{ and } a - b + c = \pi$$

$$\Rightarrow b = \pi \text{ and } a + c = 2\pi$$

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Applying  $AM \geq GM$  on  $a$  and  $c$

$$\frac{a+c}{2} \geq \sqrt{ac} \Rightarrow \pi \geq \sqrt{ac}$$

$$\Rightarrow b^2 \geq ac$$

$$\Rightarrow \frac{b^2}{ac} \geq 1 \Rightarrow \frac{ac}{b^2} \leq 1$$

**Q5 (1) - Single Correct**

$$\text{We have, } p(x) = x^2 + \frac{4x}{3} + \log_{10}(4.9) = x^2 + \frac{4x}{3} + \log_{10} 5$$

$$= x^2 + \frac{4x}{3} + 1 - \log_{10} 2$$

$$\text{For } p(x), a > 0$$

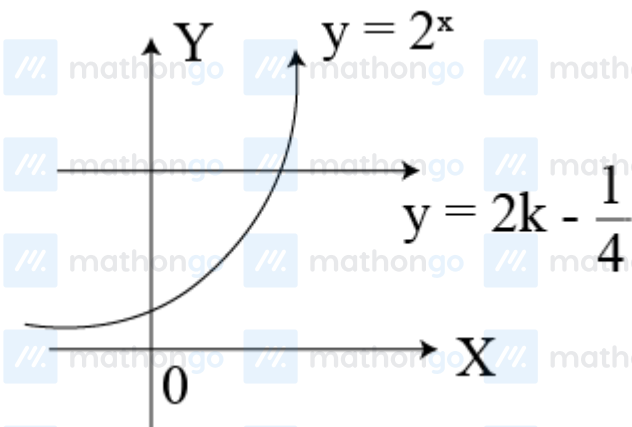
$$D = \left(\frac{4}{3}\right)^2 - 4(1 - \log_{10} 2) < 0$$

$$\Rightarrow p(x) > 0 \text{ for all } x \in R$$

$$\Rightarrow A > 0 \text{ and } B > 0$$

**Q6 (1) - Single Correct**

$$\text{We have, } 2^x = 2k - \frac{1}{4}$$



For exactly one real solution,

$$2k - \frac{1}{4} > 0 \Rightarrow k > \frac{1}{8} \text{ [as, } 2^x > 0]$$

$$\text{Hence, sum} = 1 + 2 + 3 + \dots + 100 = 5050$$

**Q7 (4) - Single Correct**

$$\text{Here, } 1 + \frac{1}{2} \sin \theta \in \left[\frac{1}{2}, \frac{3}{2}\right]$$

$$\therefore \frac{1}{2} \leq \log_3 x \leq \frac{3}{2}$$

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$$\Rightarrow \sqrt{3} \leq x \leq 3\sqrt{3}$$

$$\therefore \left| x - \frac{3}{2} \right| + \left| \frac{11}{2} - x \right| = x - \frac{3}{2} + \frac{11}{2} - x = 4$$

**Q8 (4) - Single Correct**

**Case I** When  $B$  is quadratic i.e. if  $m \neq 1$ , then

$$D_1 = (m + 3)^2 \text{ and } D_2 = (m - 2)^2$$

Roots of 1<sup>st</sup> equation are  $2, -(m + 1)$  in set  $A$ . Roots of 2<sup>nd</sup> equation are  $-1, \frac{1}{1-m}$  in set  $B$ .

For exactly 3 elements in  $A \cup B$  two of its roots must be same Possibilities are

$$2 = -(m + 1) \Rightarrow m = -3$$

$$2 = \frac{1}{1-m} \Rightarrow 2 - 2m = 1 \Rightarrow m = \frac{1}{2}$$

$$-m - 1 = -1 \Rightarrow m = 0$$

$$-(m + 1) = \frac{1}{1-m} \Rightarrow m^2 - 1 = 1 \Rightarrow m = \pm\sqrt{2}$$

$$\frac{1}{1-m} = -1 \Rightarrow m - 1 = 1 \Rightarrow m = 2$$

$$\therefore m \in \{-3, \pm\sqrt{2}, 0, 1/2, 2\} \dots (i)$$

**Case II** Now, if  $m = 1$ , then  $B$  becomes linear

Root of  $B$  is  $x = -1$  and roots of  $\Delta$  are  $2$  and  $2$  So, 3 elements in  $A \cup B$ .

$$\therefore m = \{1\} \dots (ii)$$

From Eqs. (i) and (ii),

$$m \in \{-3, \pm\sqrt{2}, 0, 1/2, 1, 2\}$$

**Q9 (4) - Single Correct**

$$\text{Here, } y(-1) = a - b + c \geq -4 \dots (i)$$

$$y(1) = a + b + c \leq 0$$

$$\text{Or } -a - b - c \geq 0 \dots (ii)$$

$$y(3) = 9a + 3b + c \geq 5 \dots (iii)$$

On adding Eqs. (i) and (ii), we get

$$-2b \geq -4 \dots (iv)$$

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On adding Eqs. (ii), (iii) and (iv), we get

$$8a \geq 1 \Rightarrow a \geq \frac{1}{8}$$

**Q10 (2) - Single Correct**

$$y^2 (5x^2 + 1) = 25 (2x^2 + 13), x, y \in 1$$

$$\Rightarrow y = \pm 5 \sqrt{\frac{2x^2+13}{5x^2+1}}$$

For integral values of  $x$  and  $y$ , we have

$$\begin{aligned} 2x^2 + 13 &= 5x^2 + 1 \\ \Rightarrow 3x^2 &= 12 \Rightarrow x = \pm 2 \end{aligned}$$

$$\text{When } x = 2, \text{ then } y = \pm 5 \sqrt{\frac{8+13}{20+1}} = \pm 5$$

When  $x = -2$ , then  $y = \pm 5$

$$2x^2 + 13 < 5x^2 + 1 \text{ for } x > 2$$

So  $\frac{2x^2+13}{5x^2+1}$  is fractional part for  $x > 2$ ,  $(2, \pm 5)$  and  $(-2, \pm 5)$  are the only 4 integrals.

**Q11 (1) - Multiple Correct**

Here,  $\alpha + \beta = -2|a|$  and  $\alpha\beta = 4$  as  $\alpha\beta \in \text{Integer}$

$$\therefore \alpha = 4, \beta = 1 \text{ or } \alpha = 1, \beta = 4$$

$$\text{or } \alpha = 2, \beta = 2$$

$$\therefore \text{Minimum value of } a = -\frac{5}{2}$$

**Q12 (3) - Multiple Correct**

$$\text{Let } f(x) = \frac{(a-b)x}{\sqrt{x^2+ax} + \sqrt{x^2+bx}} = \frac{(a-b)}{\sqrt{1+\frac{a}{x}} + \sqrt{1+\frac{b}{x}}}$$

$$\text{As, } x \rightarrow \infty f_{\max}(x) = \frac{a-b}{2}$$

$$\text{Hence, } \sqrt{x^2+ax} - \sqrt{x^2+bx} < L$$

$$L = f_{\max}(x) = 1$$

**Q13 (1,2,3,4) - Multiple Correct**

$a, b$  and  $c$  are roots of equation

$$\frac{1}{x} + \frac{1}{x-1} + \frac{1}{x-2} = 1$$

$$\Rightarrow (x-1)(x-2) + x(x-2) + x(x-1) = x(x-1)(x-2)$$

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Hints & Solutions

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$$\Rightarrow x^3 - 6x^2 + 8x - 2 = 0 \text{ has roots } a, b, c$$

$$\Rightarrow a + b + c = 6, abc = 2$$

$$\Rightarrow (1-a)(1-b)(1-c) = 1 - 6 + 8 - 2 = 1$$

$$\Rightarrow (a-2)(b-2)(c-2) = -(8 - 24 + 16 - 2) = 2$$

**Q14 (1,3) - Multiple Correct**

$$\text{Given, } x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$$

has roots  $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2 \dots$  (i)

$$\text{Now, } \frac{a}{r^2} + \frac{a}{r} + a + ar + ar^2 = 40 \dots \text{(ii)}$$

$$\text{and } \frac{r^2}{a} + \frac{r}{a} + \frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} = 10$$

On dividing Eq. (i) by Eq. (ii), we get

$$a^2 = 4 \Rightarrow a = \pm 2$$

$$\text{As, the product of roots} = a^5 = -\delta$$

$$\therefore \delta = 32 \text{ or } -32$$

**Q15 (1,3) - Multiple Correct**

$$\text{Here, } f(x) = (x^2 + 2x + 3)^2 + 2(x^2 + 2x + 3) + 3$$

$$\text{As, } x^2 + 2x + 3 > 0 \text{ for all } x \text{ } f(x) > 0 \forall x \in R$$

$$\text{Let } x^2 + 2x + 3 = y$$

$$\therefore f(x) = y^2 + 2y + 3 = (y + 1)^2 + 2$$

$$\Rightarrow f(x) = [(x^2 + 2x + 3) + 1]^2 + 2 = [(x + 1)^2 + 3]^2 + 2$$

$$\therefore f(x) \geq 9 + 2 = 11$$

**Q16 (1,2,3) - Multiple Correct**

$$x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$$

$$\Rightarrow \frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4} = \log_x \sqrt{2}$$

$$= \frac{1}{2} \log_x 2 = \frac{1}{2 \log_2 x}$$

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$$\Rightarrow 3(\log_2 x)^3 + 4(\log_2 x)^2 - 5\log_2 x - 2 = 0$$

$$\text{Let } \log_2 x = t$$

$$\Rightarrow 3t^3 + 4t^2 - 5t - 2 = 0$$

$$\Rightarrow (t - 1)(3t^2 + 7t + 2) = 0$$

$$\Rightarrow (t - 1)(3t + 1)(t + 2) = 0$$

$$\Rightarrow t = 1, t = -\frac{1}{3}, t = -2$$

$$x = 2, x = 2^{-1/3}, x = \frac{1}{4}$$

Hence, three real solutions exist and one of them is irrational.

Q17 (1,4) - Multiple Correct

Given equations are

$$x^2 + px + q = 0$$

$$\text{and } x^2 - rx + s = 0$$

$$\therefore \alpha + \beta = -p \dots (i)$$

$$\beta = q \dots (ii)$$

$$\alpha^4 + \beta^4 = r \dots (iii)$$

$$\alpha^4 \beta^4 = s \dots (iv)$$

$$q^4 = s$$

$$x^2 - 4qr + 2q^2 - r = 0$$

$$\text{Now, } D = 16q^2 - 4(2q^2 - r) = 4(2q^2 + r) \dots (v)$$

From Eq. (iii), we have

$$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

$$r + 2q^2 = (\alpha^2 + \beta^2)^2 = \text{Positive}$$

Hence, from Eq. (v),  $D > 0$ . Two real roots

Now, product of roots are

$$2(\alpha\beta)^2 - (\alpha^4 - \beta^4)$$

$$\Rightarrow -(\alpha^2 - \beta^2)^2 < 0$$

$$\Rightarrow \text{Product} < 0$$

Since, the product of roots is negative, hence one root is positive and other is negative.

Q18 (1,3) - Multiple Correct

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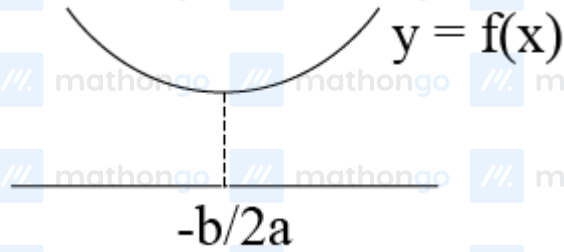
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$$f(x) = ax^2 + bx + 6 \Rightarrow f(0) = 6 > 0$$

$$\Rightarrow a > 0$$

$$\text{for } x = 3 \quad a3^2 + 3b + 6 \geq 0 \Rightarrow 3a + b \geq -2$$

$$\text{for } x = 6 \quad a6^2 + 6b + 6 \geq 0 \Rightarrow 6a + b \geq -1$$



Q19 (1,2,3,4) - Multiple Correct

$$f(0) < 0 \Rightarrow c < 0$$

Roots are real & distinct  $\Rightarrow D > 0$  product of roots  $< 0$

$$\Rightarrow \frac{c}{a} < 0 \Rightarrow a > 0$$

$$\text{Sum of roots} = -\frac{b}{a} > 0 \Rightarrow b < 0$$

Q20 (2,3,4) - Multiple Correct

$x = 1$  is a root of the equation,  $\therefore$  other root is also 1

$$\therefore \frac{ca-ab}{ab-bc} = 1$$

$$\Rightarrow 2ab = bc + ac$$

i.e  $c = \frac{2ab}{a+b}$   $\therefore a, c, b$  are in H.P.

$$\text{Since } \frac{b^2a^2c - ab^2c^2}{a^2c^2b - b^2a^2c} = \frac{ab-bc}{ac-ab} = 1$$

$\therefore ab^2c^2, b^2a^2c, a^2c^2b$  are in A.P.

Q21 (1,2) - Multiple Correct

$$\text{Let } y = x^2 + 2$$

$$y^2 + 8x^2 = 6xy$$

$$y = 4x \text{ or } 2x$$



Hints & Solutions

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if  $y = 4x$ , then  $x = 2 \pm \sqrt{2}$

If  $y = 2x$ , then there is no real solution so  $x = 2 + \sqrt{2}, 2 - \sqrt{2}$

**Q22 (1,2,3,4) - Multiple Correct**

Let the roots of the given equation be  $\alpha/r, \alpha, \alpha r$  where  $\alpha > 0$  and  $r > 1$ . Then

$$\frac{\alpha}{r} + \alpha + \alpha r = -b \dots \dots \dots (1)$$

$$\frac{\alpha}{r} \cdot \alpha + \alpha \cdot \alpha r + \frac{\alpha}{r} \cdot \alpha r = c \dots \dots \dots (2)$$

$$\text{and } \left(\frac{\alpha}{r}\right)(\alpha)(\alpha r) = 1 \dots \dots \dots (3)$$

From (3) we get  $\alpha^3 = 1$  or  $\alpha = 1$

$$\text{From (1) we get } \frac{1}{r} + 1 + r = -b \dots \dots \dots (4)$$

$$\Rightarrow \left(\frac{1}{\sqrt{r}} - \sqrt{r}\right)^2 + 3 = -b \Rightarrow -b - 3 > 0$$

$$\text{or } b < -3 = b \in (-\infty, -3)$$

$$\text{Also, from (2), } 1/r + r + 1 = c$$

$$\text{From (4) and (5) } -b = c \quad \text{or } b + c = 0$$

$$\text{As } r > 1, \alpha/r = 1/r < 1 \quad \text{and} \quad \alpha r = r > 1$$

**Q23 (2,3) - Multiple Correct**

Given equation can be Expressed as

$$\pi^e(x - \pi)(x - \pi - e) + e^\pi(x - e)(x - \pi - e) + (\pi^\pi + e^e)(x - e)(x - \pi) = 0$$

$$\text{Let } f(x) = \pi^e(x - \pi)(x - \pi - e) + e^\pi(x - e)(x - \pi - e) + (\pi^\pi + e^e)(x - e)(x - \pi)$$

$$f(e) = \pi^e(e - \pi)(-\pi) > 0$$

$$\text{and } f(\pi) = e^\pi(\pi - e)(-e) < 0;$$

hence given equation has a real root in  $(e, \pi)$

$$\text{again } f(\pi + e) = (\pi^\pi + e^e) \pi \cdot e > 0$$

$$\therefore \pi + e > \pi$$

It concludes it has a real root in  $(\pi, \pi + e)$  also

$$\therefore \pi - e < e$$

Hence  $f(x)$  has two real roots in  $(\pi - e, \pi + e)$

## Hints &amp; Solutions

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## Q24 (2) - Multiple Correct

$$x = 2 + \sqrt{3} \text{ squaring both the sides } (x - 2)^2 = 3$$

$$\text{i.e. } x^2 - 4x + 1 = 0 \quad \dots\dots\dots (i)$$

$$(x - 2)^4 = 9 \Rightarrow x^4 - 8x^3 + 24x^2 - 32x + 16 = 9$$

$$x^4 - 8x^3 + 18x^2 - 8x + 2 + 6(x^2 - 4x + 1) - 1 = 0$$

$$\text{from (i) we get } x^4 - 8x^3 + 18x^2 - 8x + 2 = 1.$$

## Q25 (3) - Multiple Correct

$$\frac{ax^2}{(1-x)^2} + \frac{bx}{1-x} + c = 0 \dots\dots\dots (i)$$

$$\text{If } t = \alpha, \text{ then } t = \frac{x}{1-x} \Rightarrow \alpha = \frac{x}{1-x}$$

$$\Rightarrow x = \frac{\alpha}{\alpha+1} \Rightarrow \text{roots of (i) are } \left( \frac{\alpha}{1+\alpha}, \frac{\beta}{1+\beta} \right)$$

## Q26 (2) - Comprehension Type

We have,

$$p(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r$$

Since,  $p(x)$  is divisible by  $x^2$ .

$$\therefore q = r = 0$$

$$\Rightarrow p(x) = x^5 - 9x^4 + px^3 - 27x^2$$

$$\frac{p(x)}{x^2} = x^3 - 9x^2 + px - 27 \text{ has roots } \alpha, \beta \text{ and } \gamma$$

$$\therefore \frac{\alpha + \beta + \gamma}{3} = 3$$

$$\therefore \text{GM of } \alpha, \beta, \gamma \Rightarrow (\alpha\beta\gamma)^{1/3} = 3$$

Since  $AM = GM$

$$\Rightarrow \alpha = \beta = \gamma = 3$$

$$\therefore p = \alpha\beta + \beta\gamma + \gamma\alpha = 27$$

$$\text{Thus, } p + q + r = 27$$

## Q27 (3) - Comprehension Type

$$\text{We have, } \alpha - 1 = 2, \beta + 3 = 6, \gamma + 7 = 10$$

$$\therefore S_n = 2 + 6 + 10 + \dots \text{ to } n \text{ terms} = 2n^2$$

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$$\begin{aligned} \text{Now, } \sum_{n=2}^{\infty} \frac{1}{\sqrt{S_n \cdot S_{n-1}}} &= \frac{1}{2} \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \frac{1}{2} \left( \frac{1}{n-1} - \frac{1}{n} \right) \\ &= \frac{1}{2} \left[ \left( 1 - \frac{1}{2} \right) + \left( \frac{1}{2} - \frac{1}{3} \right) + \left( \frac{1}{3} - \frac{1}{4} \right) + \cdots \infty \right] = \frac{1}{2} \end{aligned}$$

**Q28 (4) - Comprehension Type**

Given,  $p(0) = 2$

$\Rightarrow d = 2$

i.e. Product of all roots = 2

$\Rightarrow 1 \cdot 2 \cdot (\alpha + i\beta)(\alpha - i\beta) = 2$

$\Rightarrow \alpha^2 + \beta^2 = 1$

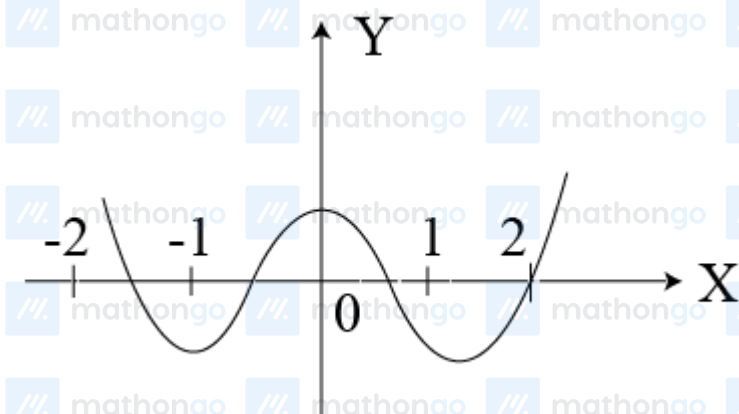
$\therefore$  Product of all imaginary roots = 1

**Q29 (1) - Comprehension Type**

Here,  $p(x) + c = 0$  has 4 real roots

$\Rightarrow c \in (-1, -2)$

$\therefore$  Graph for  $g(x) = p(x) + c$  is shown as



$\therefore -2 < \alpha < -1, -1 < \beta < 0, 0 < \gamma < 1, 2 < \delta < 3$

$\Rightarrow [\alpha] + [\beta] + [\gamma] + [\delta]$

$= -2 - 1 + 0 + 2 = -1$

**Q30 (1) - Comprehension Type**

Given,  $f(x) = x^2 - 2ax + (a^2 - 1)$

Since, both roots of Eq. (i) lies in  $(-2, 4)$ , then ... (i)

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Hints & Solutions

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$$(1) D > 0$$

$$(2) A \cdot f(-2) > 0$$

$$(3) A \cdot f(4) > 0$$

$$(4) -2 < \frac{-B}{2A} < 4$$

[ $\therefore$  if both roots of  $ax^2 + bx + c = 0$  lies in  $(k_1, k_2)$  then  $D \geq 0$ ,  
 $a \cdot f(k_1) > 0, a \cdot f(k_2) > 0$  and  $k_1 < \frac{-b}{2a} < k_2$ ]

From condition (1), we have

$$D \geq 0$$

$$\Rightarrow (-2a)^2 - 4 \times 1 \times (a^2 - 1) \geq 0$$

$$\Rightarrow 4a^2 - 4(a^2 - 1) \geq 0$$

$$\Rightarrow 4a^2 - 4a^2 + 4 \geq 0$$

$$\Rightarrow 4 \geq 0$$

$$\Rightarrow a \in R \dots (ii)$$

From condition (2), we have

$$A \cdot f(-2) > 0$$

$$\Rightarrow 1 \cdot (4 + 4a + a^2 - 1) > 0$$

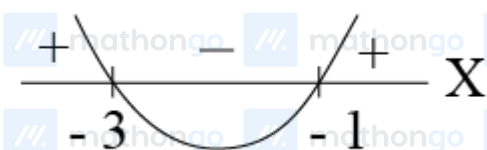
$$\Rightarrow a^2 + 4a + 3 > 0$$

$$\Rightarrow (a + 3)(a + 1) > 0$$

Equating all factors to zero, we get

$$a = -1, -3$$

By wavy curve method, we have



$$\Rightarrow a \in (-\infty, -3) \cup (-1, \infty) \dots (iii)$$

From condition (3), we have

$$A \cdot f(4) > 0$$

$$\Rightarrow 1 \cdot (16 - 8a + a^2 - 1) > 0$$

$$\Rightarrow a^2 - 8a + 15 > 0$$

$$\Rightarrow a^2 - 5a - 3a + 15 > 0$$

Hints & Solutions

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$$\Rightarrow a(a-5) - 3(a-5) > 0$$

$$\Rightarrow (a-5)(a-3) > 0$$



By wavy curve method, we have

$$\Rightarrow a \in (-\infty, 3) \cup (5, \infty) \dots \text{(iv)}$$

From condition (4), we have

$$\begin{aligned} -2 &< \frac{2a}{2} < 4 \\ \Rightarrow -2 &< a < 4 \dots \text{(v)} \end{aligned}$$

Combining Eqs. (ii), (iii), (iv) and (v), we have

$$-1 < a < 3$$

**Q31 (2) - Comprehension Type**

Since, coefficient  $x^2$ , i.e.  $4b - b^2 - 5 < 0$

Therefore, graph of  $g(x)$  is concave down.

According to the question, 1 must lie between the roots.

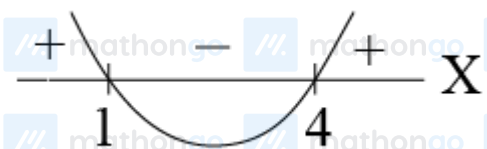
Hence,  $g(1) > 0$

$$\Rightarrow 4b - b^2 - 5 - 2b + 1 + 3b > 0$$

$$\Rightarrow -b^2 + 5b - 4 > 0$$

$$\Rightarrow b^2 - 5b + 4 < 0$$

$$\Rightarrow (b-4)(b-1) < 0$$



Equating all factors to zero, we get

$$b = 1, 4$$

$$\Rightarrow b \in (1, 4)$$

Hints & Solutions

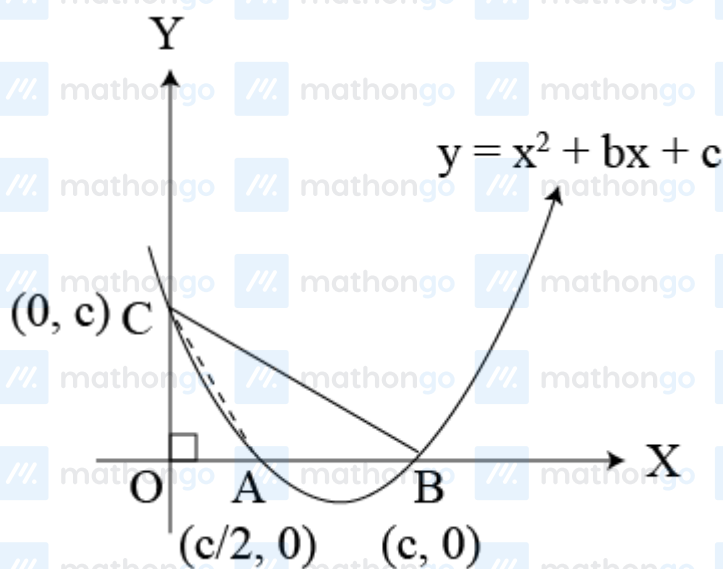
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$$\Rightarrow b = \{2, 3\}$$

Hence, two integral values of  $b$  are possible.

**Q32 (1) - Comprehension Type**

Given that,  $\triangle OBC$  is a right angled isosceles triangle in which  $AC$  is median



$$\therefore OC = OB$$

Since,  $AC$  is median, then  $A$  is mid-point of  $OB$ .

$$OA = \frac{OB}{2}$$

$$\text{Also, } y = x^2 + bx + c \dots (i)$$

Put  $x = 0$ , we get

$$y = c$$

So, the coordinate of point  $C$  is  $(0, c)$ .

$$\Rightarrow OC = c \text{ and } OB = c$$

$$\text{Then, } OA = \frac{OB}{2} = \frac{c}{2}$$

Since, curve (i) intersect  $X$ -axis at  $\left(\frac{c}{2}, 0\right)$  and  $(c, 0)$ .

So,  $\frac{c}{2}$  and  $c$  are the roots of equation  $x^2 + bx + c = 0$ .

$$\therefore \text{Sum of roots} = -b \Rightarrow \left(c + \frac{c}{2}\right) = -b$$

$$\text{and product of roots} = c \times \frac{c}{2} = c \Rightarrow \frac{c^2}{2} = c$$

$$\Rightarrow c^2 - 2c = 0 \Rightarrow c(c - 2) = 0$$

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## Hints &amp; Solutions

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$$\therefore c = 2[\because c \neq 0]$$

Hence, the roots are 1 and 2 i.e.  $\{1, 2\}$ .

**Q33 (1) - Comprehension Type**

$$\text{Here, } \alpha = 2, \beta = 1$$

$$\Rightarrow \alpha + \beta = 3, \alpha - \beta = 1$$

$$\therefore \text{Sum of roots} = \alpha + \beta + \alpha - \beta = 2\alpha = 2 \times 2 = 4$$

and product of roots = 3

Hence, required equation is  $x^2 - 4x + 3 = 0$

**Q34 (2) - Comprehension Type**

For  $a, b \in R - \{0\}$ , if  $f(x) = ax^2 + bx + a$  satisfies  $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right); \forall x \in R$

Hence,  $f(x)$  is symmetrical about  $x = \frac{7}{4}$

$$\text{i.e. } -\frac{b}{2a} = \frac{7}{4}$$

$$\Rightarrow 7a + 2b = 0 \dots (i)$$

Now,  $f(x) = 7x + a$  has only one real and distinct solution. i.e.  $ax^2 + bx + a = 7x + a$

$$ax^2 = (7 - b)x$$

$$\Rightarrow 7 - b = 0$$

$$b = 7$$

$$\Rightarrow a = -2$$

Hence,  $a + b = 5$

**Q35 (4) - Comprehension Type**

The minimum value of  $f(x)$  in  $\left[0, \frac{3}{2}\right]$  is at  $f(0)$ .

Hence, the minimum value in  $a$  is -2.

**Q36 (1) - Matrix Match (Only one option is correct)**

(i)  $ax^2 - ax + 1 > 0$  for atleast one  $x$ .

Let,  $a \geq 0$  for  $D > 0$  or  $D < 0$  So,  $ax^2 - ax + 1 > 0$  for  $a \in [0, \infty) \dots (i)$

Let,  $a < 0$ , then  $D > 0 \Rightarrow a^2 - 4a > 0$



Hints & Solutions

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$$\Rightarrow a < 0 \text{ or } a > 4$$

$$\text{So } a \in (-\infty, 0) \dots (ii)$$

$$\therefore a \in R$$

$$\therefore (i) \rightarrow (r)$$

$$(ii) f(x) = x^3 - 3x + \frac{a}{2} = 0 \text{ has 3 real and distinct root.}$$

$$\Rightarrow f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$$

$$f(1) \cdot f(-1) < 0$$

$$\Rightarrow \left(1 - 3 + \frac{a}{2}\right) \left(-1 + 3 + \frac{a}{2}\right) < 0$$

$$\Rightarrow \left(\frac{a}{2} - 2\right) \left(\frac{a}{2} + 2\right) < 0$$

$$\Rightarrow 0 \leq |a| < 4$$

$$\therefore (ii) \rightarrow (p)$$



$$(iii) \text{ Here, } f(x) = x^3 + ax^2 + ax + 1 = 0 \text{ is an increasing function.}$$

$$\therefore f'(x) = 3x^2 + 2ax + a \geq 0$$

$$\Rightarrow 4a^2 - 12a \leq 0$$

$$\Rightarrow 4a(a - 3) \leq 0$$

$$0 \leq a \leq 3$$

$$\therefore (iii) \rightarrow (q)$$

$$(iv) \text{ Here, } x^2 - 3ax + (a^2 - 9) = 0 \text{ has roots of opposite sign, then product of roots } < 0$$

$$\Rightarrow a^2 - 9 < 0 \Rightarrow a \in (-3, 3)$$

$$\therefore (iv) \rightarrow (s)$$

**Q37 (1) - Matrix Match (Only one option is correct)**

Here,  $x^3 + ax^2 + bx + c = 0$  has 3 distinct integral roots and  $[g(x)]^3 + a[g(x)]^2 + b[g(x)] + c = 0$  has no real roots, where  $g(x) = x^2 + 2x + 2$

Let the roots of  $x^3 + ax^2 + bx + c = 0$  be  $x_1 > x_2 > x_3$  respectively.

Since,  $g(x)$  can take all values from  $[1, \infty)$ .

Hints & Solutions

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$$\Rightarrow x_1 \leq 0, x_2 \leq -1, x_3 \leq -2$$

$$\therefore a = -(x_1 + x_2 + x_3) \geq 3$$

$$b = x_1x_2 + x_2x_3 + x_3x_1$$

$$\Rightarrow b \geq 2$$

$$\text{and } c = -(x_1x_2x_3) \Rightarrow c \geq 0$$

Since, roots of  $f'(x) = k$  are equal.

So,  $3x^2 + 2ax + (b - k) = 0$  has equal roots

$$\Rightarrow a^2 = 3(b - k) \Rightarrow k = b - \frac{a^2}{3} = 2 - 3 = -1$$

$$\therefore (i) \rightarrow r, (ii) \rightarrow q_1, (iii) \rightarrow p, (iv) \rightarrow s$$

**Q38 (1) - Matrix Match (Only one option is correct)**

(i) We have,  $y = ax^2 + bx + c \dots(i)$

where,  $a, b, c \in R, a \neq 0$

Here,  $a = 1, c = 4$

It is given that, both roots of Eq. (i) are greater than 2.

$$(1) D \geq 0$$

$$(2) a \cdot f(2) > 0$$

$$(3) 2 < \frac{-b}{2a}$$

From condition (1), we have

$$D \geq 0$$

$$\Rightarrow b^2 - 4ac \geq 0$$

$$\Rightarrow b^2 - 4 \times 1 \times 4 \geq 0$$

$$\Rightarrow b^2 - 16 \geq 0$$

$$\Rightarrow (b - 4)(b + 4) \geq 0$$

Equating all factors to zero, we get

$$\Rightarrow b = -4, 4$$

$$\Rightarrow b \in (-\infty, -4) \cup (4, \infty) \dots(ii)$$



From condition (2), we have

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Hints & Solutions

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$$a \cdot f(2) > 0$$

$$\Rightarrow 1 \cdot (4 + 2b + 4) > 0$$

$$\Rightarrow 2b + 8 > 0$$

$$\Rightarrow b > -4$$

From condition (3), we have

$$\Rightarrow 2 < \frac{-b}{2a} \Rightarrow 2 < \frac{-b}{2}$$

$$\Rightarrow 4 < -b \Rightarrow -4 > b$$

$$\Rightarrow b < -4$$

Combining Eqs. (ii), (iii) and (iv), we get

$$b \in \phi$$

Hence,  $b$  has no values.

$$\therefore (i) \rightarrow (s)$$

$$(ii) \text{ We have, } y = ax^2 + bx + c \dots(i)$$

$$\text{Here, } a = -1, b = 5$$

It is given that ' -1 ' lies between the roots of Eq. (i).

$$(1) D > 0 (2) a f(-1) < 0$$

From condition (1), we have

$$D > 0$$

$$\Rightarrow b^2 - 4ac > 0$$

$$\Rightarrow (5)^2 - 4(-1)c > 0$$

$$\Rightarrow 25 + 4C > 0$$

$$\Rightarrow 4c > -25$$

$$\Rightarrow c > -\frac{25}{4} \dots(ii)$$

From condition (2), we have

$$a f(-1) < 0$$

$$\Rightarrow -1 f(-1) < 0$$

$$\Rightarrow -(-1 - 5 + C) < 0$$

$$\Rightarrow -(C - 6) < 0$$

$$\Rightarrow C - 6 > 0 \dots(iii)$$

$$\Rightarrow c > 6$$

From Eqs. (ii) and (iii), we get

Hints & Solutions

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$$C \in (6, \infty)$$

$$(ii) \rightarrow (q, r)$$

$$(iii) \text{ Given, } y = ax^2 + bx + c$$

$$\text{Here } b = 6, c = 1$$

It is given that one root of Eq. (i) is less than -1 and other root greater than  $-\frac{1}{2}$ , then

$$(1) D > 0$$

$$(2) af(-1) < 0$$

$$(3) a f\left(-\frac{1}{2}\right) < 0$$

[ $\because$  if two numbers  $k_1$  and  $k_2$  lie between the roots of  $ax^2 + bx + c = 0$ , then

$$D > 0, af(k_1) < 0, af(k_2) < 0]$$

From condition (1), we have

$$D > 0$$

$$\Rightarrow b^2 - 4ac > 0$$

$$\Rightarrow (6)^2 - 4a \times 1 > 0$$

$$\Rightarrow 36 - 4a > 0$$

$$\Rightarrow 36 > 4a$$

$$\Rightarrow 9 > a$$

$$\Rightarrow a < 9 \dots (ii)$$

From condition (2), we have

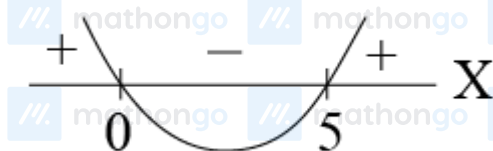
$$af(-1) < 0$$

$$\Rightarrow a(a - 6 + 1) < 0$$

$$\Rightarrow a(a - 5) < 0$$

Equating all factors to zero, we get

$$a = 0, 5$$



$$\Rightarrow a \in (0, 5)$$

From condition (3), we have

Hints & Solutions

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$\Rightarrow$

$$a f\left(-\frac{1}{2}\right) < 0$$

$$a\left(\frac{a}{4} - 3 + 1\right) < 0$$

$$\Rightarrow a\left(\frac{a}{4} - 2\right) < 0$$

$$\Rightarrow a(a - 8) < 0$$

Equating all factors to zero, we get

$$a = 0, 8$$



$$\Rightarrow a \in (0, 8) \dots \text{(iv)}$$

Combining Eqs. (ii), (iii) and (iv), we get

$$a \in (0, 5)$$

Hence, (iii)  $\rightarrow$  (p)

**Q39 (2) - Integer Type**

roots of  $x^2 - |a - 1|x = 0$  are  $x = 0$  and  $|a - 1|$

It should be equal to  $|x| - 1$

So, when  $x < 0$  equation should have 1 real root

$$x^2 - |a - 1|x + 1 = -x$$

$$x^2 - x(|a - 1| - 1) + 1 = 0$$

$$D = 0$$

$$|a - 1|^2 - 2|a - 1| - 3 = 0$$

$$|a - 1| = 3$$

$$a = 4, -2$$

when  $x > 0$ , equation should have 2 real roots

$$x^2 - |a - 1|x + 1 = x$$

$$x^2 - (|a - 1| + 1)x + 1 = 0$$

Hints & Solutions

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$$D > 0$$

$$(|a - 1| + 3)(|a - 1| - 1) > 0$$

$$|a - 1| > 1$$

$$\text{so, } a > 2 \text{ or } a < 0$$

Take intersection of both cases

$$a = 4, -2$$

$$\sin = 2$$

**Q40 (1) - Integer Type**

$$\text{Here, } \frac{x(x^6+1)}{x^2+1} = a$$

**Case I** If  $x > 0$ , then  $a > 0$

$$\text{So, } x^6 + 1 = a \left( \frac{x^2+1}{x} \right) = a \left( x + \frac{1}{x} \right) \geq 2a$$

$$\therefore x^6 \geq 2a - 1$$

**Case II** If  $x < 0$ , then  $a < 0$

$$x^6 + 1 = a \left( x + \frac{1}{x} \right) \geq 2a$$

$$x^6 \geq 2a - 1$$

**Q41 (6) - Integer Type**

$$\text{Here, } |x^2 - (7 + k^2)x + 7k^2| + \sqrt{(x-3)(x-3k+2)} = 0$$

$$\Rightarrow |(x-7)(x-k^2)| + \sqrt{(x-3)(x-(3k-2))} = 0$$

The above equation has a solution, if equation  $(x-7)(x-k^2) = 0$  and  $(x-3)(x-(3k-2)) = 0$  has atleast one common root.

$$\therefore k^2 = 3 \Rightarrow k = \pm\sqrt{3}$$

$$\text{or } k^2 = 3k - 2 \Rightarrow k^2 - 3k + 2 = 0$$

$$\Rightarrow k = 1 \text{ or } 2 \text{ and } 3k - 2 = 7 \Rightarrow k = 3$$

$$\therefore \text{Sum of all values of } k = \sqrt{3} - \sqrt{3} + 1 + 2 + 3 = 6$$

**Q42 (5) - Integer Type**

Hints & Solutions

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Here,  $3x^2 + 2x - 1 < 0$

$\Rightarrow (x+1)(3x-1) < 0$

So,  $(3a-2-a^2)x+2 < 0$  for some  $x$  in  $(-1, \frac{1}{3})$  and  $(a^2-3a+2)x-2 > 0$  for some  $x$  in  $(-1, \frac{1}{3})$



Now, Let  $f(x) = (a^2 - 3a + 2)x - 2$

$\therefore f(-1) = -a^2 + 3a - 4 < 0$

and  $f(\frac{1}{3}) = \frac{(a^2-3a+2)}{3} - 2 > 0$

$\Rightarrow a^2 - 3a + 4 > 0$



and  $a^2 - 3a - 4 > 0$

$\Rightarrow a^2 - 3a - 4 > 0$

$\Rightarrow (a-4)(a+1) > 0$

$\therefore a < -1$  or  $a > 4$

Thus, least natural number is 5.

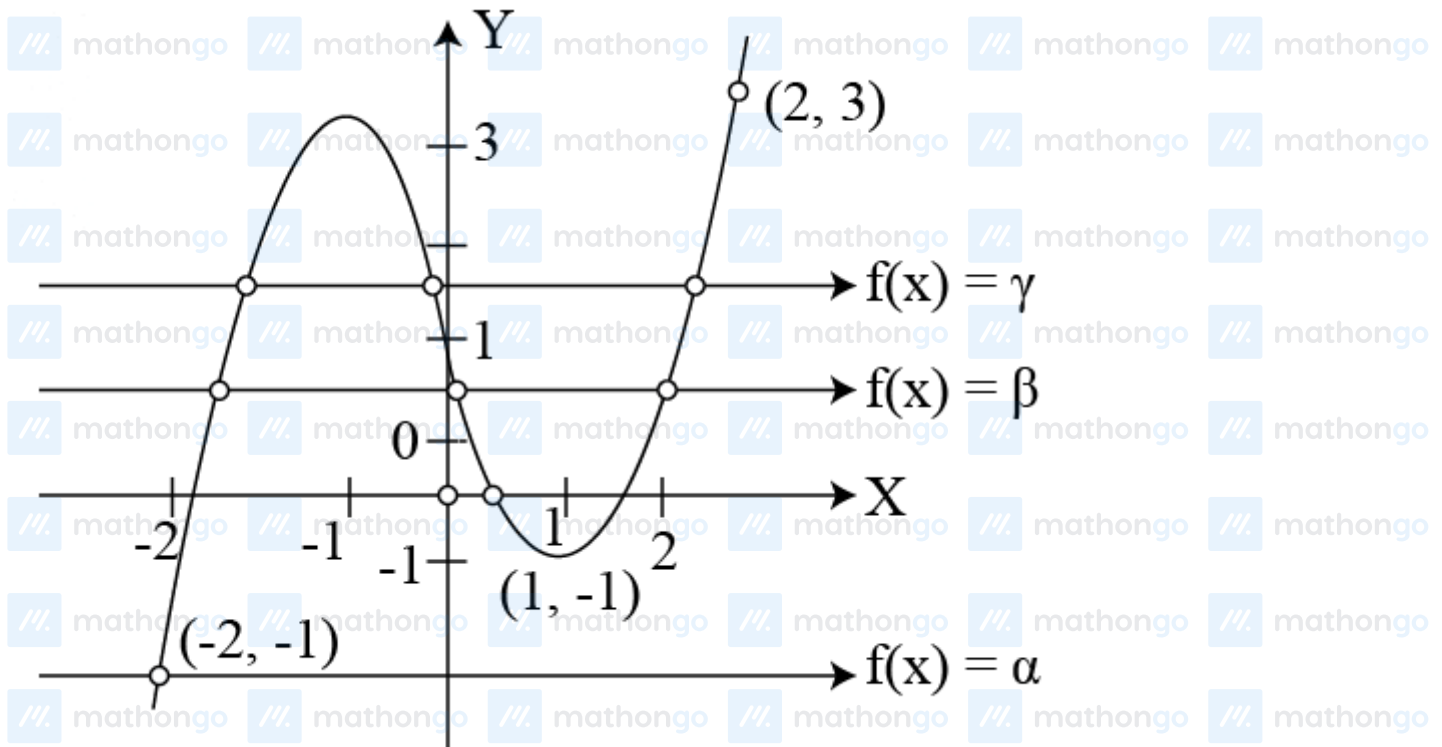
Q43 (7) - Integer Type



Hints & Solutions

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The equation  $f(x) = 0$  has 3 distinct real roots  $\alpha, \beta, \gamma$  where  $-2 < \alpha < -1, 0 < \beta < 1$  and  $1 < \gamma < 2$



So, the equation  $f(f(x)) = 0$  has 7 solutions.

**Q44 (0) - Integer Type**

Let,  $x_1, x_2, x_3, x_4$  be the real roots of the polynomial  $f$ .

Then,  $f = (x - x_1)(x - x_2)(x - x_3)(x - x_4)$

$$\text{and } |f(i)| = \sqrt{1 + x_1^2} \sqrt{1 + x_2^2} \sqrt{1 + x_3^2} \sqrt{1 + x_4^2} \dots (i)$$

$$\text{But } |f(i)| = 1 \dots (ii)$$

From Eqs. (i) and (ii) we get

$$x_1 = x_2 = x_3 = x_4 = 0$$

$$\therefore a = b = c = d = 0$$

$$\Rightarrow a + b + c + d = 0$$

**Q45 (2) - Integer Type**

Given equation,  $x^2 - 8kx + 16(k^2 - k + 1) = 0$

Since, roots are real and distinct and have values atleast 4

Hints & Solutions

#MathBoleTohMathonGo

(1)  $D > 0$  (2)  $f(4) \geq 0$  (3) Vertex of parabola to the as right of  $(4, 0)$ , i.e.  $-\frac{b}{2a} > 4$ .

From condition (1), we have

$$\Rightarrow (-8k)^2 - 4 \times 1 \times 16(k^2 - k + 1) > 0$$

$$\Rightarrow 64k^2 - 64(k^2 - k + 1) > 0$$

$$\Rightarrow 64(k^2 - k^2 + k - 1) > 0$$

$$\Rightarrow 64(k - 1) > 0$$

$$\Rightarrow k - 1 > 0$$

$\therefore 64 > 0$ , if  $ab > 0$  and  $a > 0 \Rightarrow b > 0$

$$\Rightarrow k > 1$$

$$\Rightarrow k \in (1, \infty) \dots (i)$$



From condition (2), we have

$$16 - 32k + 16(k^2 - k + 1) \geq 0$$

$$\Rightarrow k^2 - 3k + 2 \geq 0$$

$$\Rightarrow (k - 1)(k - 2) \geq 0$$

Equating all factors to zero, we get

$$k = 1, 2$$

$$\Rightarrow k \in (-\infty, 1] \cup [2, \infty) \dots (ii)$$

From condition (3), we have

$x$ -coordinate of vertex  $> 4$

$$\text{i.e. } -\frac{b}{2a} > 4 \Rightarrow 4k > 1$$

$$\Rightarrow k > 1 \dots (iii)$$

Combining Eqs. (i), (ii) and (iii), we get

$$k \in [2, \infty)$$

Hence, the smallest value of  $k$  is 2.

Q46 (4) - Integer Type

Hints & Solutions

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Given equation

$$x^2 - (a - 5)x + \left(a - \frac{15}{4}\right) = 0 \dots (i)$$

If one root of Eq. (i) greater than zero. Then.

$$(1) D \geq 0 (2) a \cdot f(0) < 0$$

From condition (1), we have

$$D \geq 0$$

$$\Rightarrow (a - 5)^2 - 4\left(a - \frac{15}{4}\right) \geq 0$$

$$\Rightarrow (a^2 + 25 - 10a) - 4a + 15 \geq 0$$

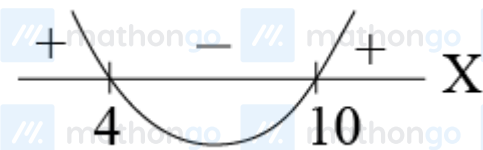
$$\Rightarrow a^2 + 25 - 10a - 4a + 15 \geq 0$$

$$\Rightarrow a^2 - 14a + 40 \geq 0$$

$$\Rightarrow (a - 10)(a - 4) \geq 0$$

Equating all factors to zero, we get

$$a = 4, 10$$



$$\Rightarrow a \in (-\infty, 4] \cup [10, \infty) \dots (ii)$$

From condition (2), we have

$$a \cdot f(0) < 0$$

$$\Rightarrow a - \frac{15}{4} < 0$$

$$\Rightarrow a < \frac{15}{4}$$

$$\Rightarrow a < 3.75 \dots (iii)$$

Combining Eqs. (ii) and (iii), we get

$$a \in \left(-\infty, \frac{15}{4}\right) \dots (iv)$$

Now if both roots of Eq. (i) are greater than zero, then

$$(1) a \cdot f(0) > 0 (2) D \geq 0$$

From condition (1), we get

Hints & Solutions

#MathBoleTohMathonGo

$$a - \frac{15}{4} > 0 \Rightarrow a > \frac{15}{4}$$

$$\Rightarrow a \in \left( \frac{15}{4}, \infty \right) \dots (v)$$

From condition (2), we have

$$D \geq 0$$

$$(a - 10)(a - 4) \geq 0$$

$$\Rightarrow a \in (-\infty, 4] \cup [10, \infty) \dots (vi)$$

[For eqn. (ii)]

Combining Eqs. (v) and (vi), we get

$$a \in [10, \infty) \dots (vii)$$

Hence, if atleast one root of eqn. (i) is greater then zero then (iv) and (vii) hold good.

$$\text{i.e. } a \in \left( -\infty, \frac{15}{4} \right) \cup [10, \infty)$$

So, the integral values of a in the interval [1,100] are

$\therefore$  Sum of all integral values of 'a'

$$= (1 + 2 + 3) + (10 + 11 + 12 + 13 + \dots + 100)$$

$$= (1 + 2 + 3) + (10 + 1) + 100 - (1 + 2 + 3 + \dots + 9)\}$$

$$= 6 + \frac{100 \times 101}{2} - \frac{9 \times 10}{2} = 6 + 5050 - 45 = 5011$$

Number of digits in 5011 is 4.

**Q47 (1) - Integer Type**

Given equation,  $x^2 - kx + k^2 + 7k + 15 = 0 \dots (i)$

Since,  $x_1$  and  $x_2$  are real roots of Eq. (i).

Then.  $D \geq 0$

$$\Rightarrow (-k)^2 - 4(k^2 + 7k + 15) \geq 0$$

$$\Rightarrow k^2 - 4k^2 - 28k - 60 \geq 0$$

$$\Rightarrow -3k^2 - 28k - 60 \geq 0$$

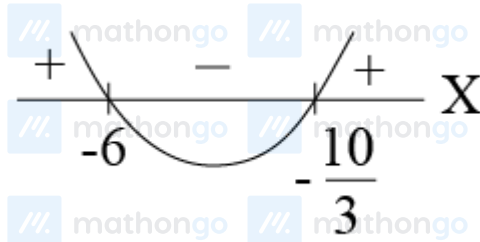
$$\Rightarrow -(3k^2 + 28k + 60) \geq 0$$

$$\Rightarrow 3k^2 + 28k + 60 \leq 0$$

$$\Rightarrow (k + 6)(3k + 10) \leq 0$$

Hints & Solutions

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Equating all factors to zero, we get

$$k = -6, -\frac{10}{3}$$

$$\Rightarrow k \in \left[-6, -\frac{10}{3}\right] \dots \text{(ii)}$$

Also,  $x_1$  and  $x_2$  are roots of Eq. (i).

$$\therefore x_1 + x_2 = k \dots \text{(iii)}$$

$$x_1 \cdot x_2 = k^2 + 7k + 15 \dots \text{(iv)}$$

$$\text{Now, } x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2$$

$$= (k)^2 - 2(k^2 + 7k + 15)$$

$$= k^2 - 2k^2 - 14k - 30 = -k^2 - 14k - 30$$

$$\Rightarrow x_1^2 + x_2^2 = -(k^2 + 14k + 30) = -[k^2 + 14k + 49 + 30 - 49]$$

$$= -[k^2 + 14k + (7)^2 - 19] = -[(k + 7)^2 - 19]$$

$$\Rightarrow x_1^2 + x_2^2 = 19 - (k + 7)^2 \dots \text{(v)}$$

Now,  $x_1^2 + x_2^2$  will be maximum, when  $k + 7$  is zero

$$\text{ie. } k + 7 = 0$$

$$k = -7$$

But from Eq. (ii),  $k$  cannot be -7.  $\left[\because k \in \left[-6, -\frac{10}{3}\right]\right]$

$\therefore$  Minimum value of  $k = -6$

$$\text{Maximum value of } x_1^2 + x_2^2 = 19 - (-6 + 7)^2 = 19 - 1 = 18$$

But maximum value  $\frac{18}{x}$

$\therefore$  Value of  $x = 1$

**Q48 (0) - Integer Type**

Hints & Solutions

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Given inequality  $x^2 - 14x + 40 \leq 0$

$$(x - 10)(x - 4) \leq 0$$

Hence,  $x \in [4, 10]$

Now, other given inequality

$$x^2 - 6ax + 5a^2 \leq 0$$

$$\Rightarrow (x - a)(x - 5a) \leq 0$$

Hence,  $x \in [a, 5a]$

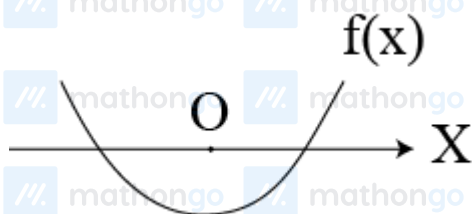
Since, in the integers  $x \in [4, 10]$ , none of the integer is a multiple of 5 of the other integer.

Hence, there are zero possible integral value of  $a$  for which both the inequalities have same solution.

Q49 (5) - Integer Type

Let  $f(x) = x^2 - 2(a + 1)x + 3(a - 3)(a + 1)$

Case I

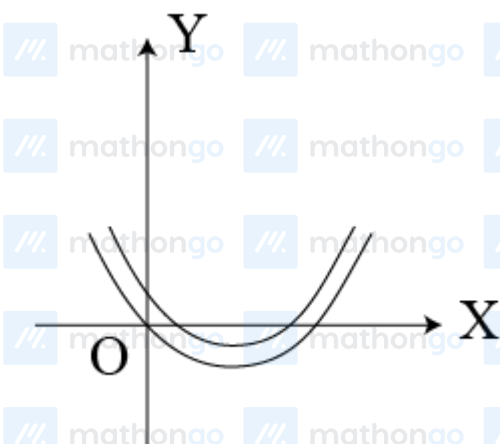


$$f(0) < 0$$

$$\Rightarrow (a - 3)(a + 1) < 0$$

$$\Rightarrow -1 < a < 3$$

Case II



Hints & Solutions

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$$(i) f(0) \geq 0$$

$$\Rightarrow (a-3)(a+1) \geq 0$$

$$\Rightarrow a \in (-\infty, -1] \cup [3, \infty)$$

$$(ii) D > 0$$

$$\Rightarrow 4[(a+1)^2 - 3(a-3)(a+1)] > 0$$

$$\Rightarrow a^2 + 2a + 1 - 3(a^2 - 2a - 3) > 0$$

$$\Rightarrow -2a^2 + 8a + 10 > 0$$

$$\Rightarrow a^2 - 4a - 5 < 0$$

$$\Rightarrow (a-5)(a+1) < 0$$

$$\Rightarrow a \in (-1, 5)$$

$$(iii) -\frac{B}{2A} > 0 \Rightarrow (a+1) > 0 \Rightarrow a > -1$$

From part (i)  $\cap$  part (ii)  $\cap$  part (iii), we get

$$a \in [3, 5) \dots (ii)$$

Case I  $\cup$  Case II

$$\Rightarrow a \in (-1, 3) \cup [3, 5)$$

So, there are 5 possible integral values of  $a$

i.e.  $a = 0, 1, 2, 3, 4$

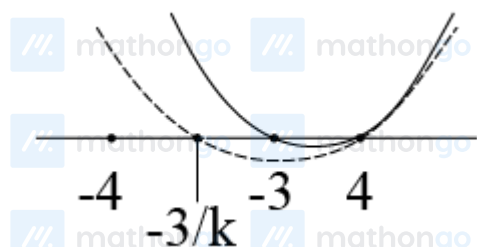
Q50 (6) - Integer Type

$$f(x) = (kx+3)(x-4)$$

$$\Rightarrow f(x) < 0 \text{ for } x \in (-3, 3) \text{ and } f(-4) > 0$$

$$-4 < \frac{-3}{k} \leq -3$$

$$\Rightarrow 4 > \frac{3}{k} \geq 3$$



$$\Rightarrow \frac{1}{4} < \frac{k}{3} \leq \frac{1}{3}$$

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Hints & Solutions

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|                                                |             |             |             |             |             |             |
|------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| $\Rightarrow \frac{3}{4} < k \leq 1$           | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo |
| $\therefore k \in \left(\frac{3}{4}, 1\right]$ | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo |
| $\therefore p = \frac{3}{4}$ and $q = 1$       | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo |
| $\Rightarrow 4p + 3q = 6$                      | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo | // mathongo |
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